

ABOUT SENSITIVITY ANALYSIS IN LP (LINEAR PROGRAMMING) MODELS

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Introduction and Summary.

The following formulation is said to have been coined by P.Rivett, the holder of the first chair of Operations Research in Britain: "One job of the Operations Research worker is to look at the problem proposed and say: Did this really matter to the Company, or were they carrying out some ritual algebraic dance around the decision point? If so, let them give it up.

A typical example of a proposed problem that might, after scrutiny, turn out to be a ritual algebraic dance is any (particularly any large scale) LP model, ^(*) as long as no error analysis (or sensitivity analysis in the terminology of Dantzig) has been made.

The topic of the present paper is this scrutiny for the influence of coefficient errors only. Any other possible objections to LP models (as e.g. those concerning assumptions of linearity) are not considered i.e. are assumed to be non existent for the purpose of the present paper.

Every engineer knows (and uses this knowledge) that in spite of the correctness of all his formulae he has to be very careful to make sufficient allowances for discrepancies between the assumed model and the restricted accuracy of available knowledge.

It is strange that many "applied" mathematicians, of all people, should display a kind of pious belief that in the case of LP models error considerations may safely be neglected. This pious belief is often disguised in the following pseudo logical argument:-

- (a) If coefficient errors cause a change of basis this may obviously introduce large and discontinuous changes of the variables (which is true).
- (b) But: The object function is continuous in spite of the discontinuities of the variables (which is also true apart from the borderline case of an unbounded object function).
- (c) "Hence" (which is the wrong conclusion) it cannot be too bad.

It is intended in the present paper to do the following three things:-

- (1) To show by an example (in concrete figures) that the "con-

^{*)} See Dantzig's famous textbook p. 265 and particularly p. 267 top (c).

clusion" under (c) above is wrong, since one may (as in the example) construct cases where it is extremely bad. Hence one cannot claim without a closer look at each particular case that it is "not too bad".

It may be considered superfluous to disprove this kind of obvious fallacy by an example; the only justification is that the author has heard the argument put forward in discussions by several different mathematicians as long as the example was not available.

(2) To report about presently available methods of dealing with the error problem.

This leads to the deplorable result that there is no worked out general solution.

(3) To suggest as a substitute for a proper error analysis some crude trial procedure which at least has a chance of detecting a ritual dance in some cases and hence of increasing the subjective confidence of the user of the model in the others.

In addition to this, there is an overcritical estimate of boundaries of the object function which may be useful in very safe cases.

Summing up

It is claimed that a responsible mathematician cannot suggest to use an LP model naively, i.e. without sensitivity analysis. This is so because one would not know in any particular case whether the result produced by the computer makes sense in practice.

Unfortunately, a proper error analysis is not available; still less is it feasible for large models.

Hence for a mathematician there is hardly any choice left than that between (A) and (B) below:

(A) To admit that he does not know and hence cannot suggest the use of large models.

(B) To suggest some substitute for a proper error analysis.

This choice between (A) and (B) applies to a mathematician in his own right, not to defend his or anyone else's financial interests, but to say what he considers reasonable; whereas a management consultant or a salesman of computers cannot be blamed for advocating the naive use of any (maybe even preferably large) LP models.

The fact that it is usually easy enough for a mathematician to hide away the difficulty because non-mathematicians are often used to accept his solution as gospel truth is no excuse for his suggesting the naive use of LP models.

An Example of Large Induced Errors of the Object Function

b	-18.457	25.900	22.946	0	0	$-\infty$	0	0	0
Y_0	Y_1	Y_2	Y_3	Y_4	T_1	T_2	T_3	T_4	T_5
6	2.8235	-3.1176	-2.8824	0	1				
10	-3.8009	6.6968	6.3801	-1		1			
30	.2489	-2.6290	4.4751	0			1		
18	α	β	γ	0				1	
0	.0588	.7059	-.7059	0					1

To maximise $B = \Sigma$ by

The only parameters are α , β and γ , for which 3 values each were chosen and all 27 combinations used. All other coefficients are fixed constants.

	α	β	γ	The result is:
1	-9.4	13.3	12.2	
2	-10.7	14.3	13.3	$B(\alpha_1 \beta_3 \gamma_3) = 11.0 \leq B \leq 125.5 = B(\alpha_3 \beta_1 \gamma_1)$
3	-12.0	15.3	14.4	

The cube produced in the parameter space is subdivided into 6 different regions of basis. Any particular value B computed would be a useless estimate for many of the other 26 points considered.

In particular the centre point $B(\alpha_2 \beta_2 \gamma_2) = 40$ is surrounded by the following 6 values of B , each involving a change of one single parameter and a change of basis:-

$B = \underbrace{28, 30, 31}_{\text{common basis}},$	$\underbrace{57, 62, 90}_{\text{common basis}}$
common basis	common basis

An example of how misleading an estimate using the derivative of B at an adjacent point may be is:-

$$B(\alpha_1 \beta_1 \gamma_1) = 34.1 \quad B(\alpha_2 \beta_1 \gamma_1) = 77.9$$

$$\frac{\partial B}{\partial \alpha} = -1.61; \quad \partial_{\alpha} = -1.3$$

estimate $\partial B \approx 2.1$ whereas $\Delta B = 43.8$

There is a change of basis between the two points.

Report on Available Methods

A proper error analysis would have to contain:

- I A procedure to estimate the influence of a change of a single coefficient a_{ij} to $a_{ij} + \delta$ on the object function $C = C(\text{Min})$.

II. A procedure to estimate the combined influence of various δ of all a_{ij} on C .

To make a working tool of it, it would further have to contain:

III I and III II, programme facilities on a computer for application of I and II respectively.

The present position is:

I is available.

Depending on the particular model more or less of III I is available on the computer in various subroutines. (More on more modern and larger models).

A complete programme to be used in a routine way does not exist anywhere at present.

II has not been worked out so far*) and is rather a hard nut to crack. Hence it would be unduly optimistic to expect anything like III II to be available in the foreseeable future for numerical work on actual problems.

I Estimation of the Influence of a Change of one single a_{ij} to $a_{ij} + \delta$ on C .

Since the knowledge of I alone does not put one in an essentially improved position with respect to the required III II, the pure facts are mentioned subsequently without going into details of proof.

I immediately breaks down into two parts:

I.1. Influence of δ on C assuming that the basis is unchanged.

I.2. Influence of δ on C if δ caused a change of basis.

I.1. Basis unchanged

This case is pretty simple. Writing the LP model as
 $Ax = b$ excluding nonbasic variables

$$c'x = C \text{ (Min)}$$

one obtains by total differentiation

$$dC = c' A^{-1} db + (dc' - c' A^{-1} dA)x$$

In particular if $db = dc = 0$

$$dC = -c' A^{-1} dAx \text{ or, as this is usually written}$$

*) A remark by Dantzig (textbook p. 265) "The reader will find no difficulty in extending the results to the general case" is probably meant as a challenge to solve only I above, not II.

$\frac{\partial C}{\partial a_{ij}} = -h_s x_j$ where h_s is the shadow price of the slack variable s_i = with

coefficient -1 in row i originally and x_j is the value of the variable attached to a_{ij} produced in the computer's solution.

In this formula the only case of interest is:

x_j is basic (x_j is B) } in both the primal and
 s_i is nonbasic (s_i is N) } (hence also in the) dual problems.

In all other cases (some of which were excluded by the above notation) the formula is a triviality and hence it is generally correct.

If it can be assumed that $|\delta|$ is sufficiently small one may safely use

$\Delta C \approx -h_s x_j \delta$ as an estimate of the change

induced by δ as required.

An error in a coefficient attached to an N does not influence C at all as long as the variable does not enter the basis.

If $|\delta|$ is appreciable the estimate of ΔC produced may become bad owing to a quick change of the derivative. But this is still harmless compared with an attempted application of the formula to cases where the basis changes. Here, even with small $|\delta|$, the estimate may become completely useless due to discontinuity of the derivative.

I.2. Change of Basis

The main question is: Given the LP problem, involving a large matrix, and its solution produced by the computer; how does one recognise whether or not a change of a_{ij} to $a_{ij} + \delta$ produces a change of basis?

For a given value of δ one could "obviously" put the changed matrix to test by producing a second solution on the computer. The question is: How to avoid this additional computation in a general and simpler procedure? Such a general procedure exists.*)

The kind of influence of δ on changes of basis depends on whether the variables x_j and s_i (or $e_i = -s_i$) are (separately) B or N.

It is convenient to put the four possible cases like this:

Primal		Dual		
x_j	s_i	t_j	y_i	
1. B	B	N	N	1. dual to 4.
2. B	N	N	B	2. self dual
3. N	B	B	N	3. self dual
4. N	N	B	B	4. dual to 1.

*) I am indebted to Mr. E.M.L. Beale of CEIR London for the knowledge of it.

where t_j is the slack in the dual corresponding to x_j , and y_i is the dual variable corresponding to s_i .

Since one may always switch to the dual and/or back without undue inconvenience on a large computer, preference of a step in the dual to the corresponding one in the primal or vice-versa is unnecessary, apart from expediency of the steps themselves. Hence 1. and 4. are equivalent and it is sufficient to consider e.g. 2. 3. and 4. only.

3. and 4. are simpler than 2. They have in common that x_j is N.

Hence the Matrix A (the check matrix bringing back the final tableau to the original form) does not contain the vector a_j (in column j of the original tableau); therefore the final tableau cannot contain δ anywhere but in column j. Now δ (as a change of a_j) behaves during the simplex computation proportionally to the column of s_i (since x_j is N). Hence δ cannot influence any of the basic variables and its only influence on the solution is restricted to h_j , the shadowprice of x_j . The change of h_j produced by δ is

$$h_j(0) \text{ changes to } h_j(\delta) = h_j(0) - \delta h_s$$

If s_i is B then $h_s = 0$ and δ has no influence at all. This solves completely (and trivially) the self dual case 3.

If the change produces a negative $h_j(\delta)$ a change of basis has occurred at the critical δ_0 where

$$\delta_0 = \frac{h_j(0)}{h_s}$$

If such a change of basis occurs one has to repeat the whole procedure (it will be a case 2. now since x_j has entered the basis).

Case 2. x_j is B; s_i is N. Still a_{ij} is changed to $a_{ij} + \delta$.

The influence of δ is spread all over the final tableau. One needs some means of changing the tableau to a state where δ appears only in a restricted number of places in order to avoid repeating the whole simplex procedure changing the original matrix by a single δ . If one introduces s_i (assumed N) artificially into the basis instead of x_j , regardless of signs of variables and changes of C in the unfavourable direction, δ must disappear from everywhere but column j, because x_j is now N. Since s_i is now B and has its unit element in the row where x_j had its unit before the change (say row p) δ appears only in a single place; i.e. the coefficient in row p column j of the artificial tableau is $a - \delta$ if it was originally a.

This is a useful result. Using it we can compute the overall influence of δ on the final tableau applying one change of basis and its inverse after changing the single coefficient a to $a - \delta$ in-between, instead of recomputing the whole simplex procedure.

The result can be expressed in a simple set of formulae which can be computed by hand in general and incorporated in the solution programme. Work is proceeding along these lines but the programme is not yet available. *)

The formulae are:

If one denotes by $(\alpha\beta)(\delta)$ the entry in the final tableau in row α column β , putting $(\alpha\beta)(0) = (\alpha\beta)$ one obtains:

$$(\alpha\beta)(\delta) = \frac{(\alpha\beta) - D\delta}{k} \quad \text{or} \quad \frac{\Delta(\alpha\beta)}{\delta} = \frac{(\alpha s)(p\beta)}{k}$$

$$\text{where } k = 1 - (ps) \cdot \delta ; D = \begin{vmatrix} (\alpha\beta) & (\alpha s) \\ (p\beta) & (ps) \end{vmatrix} ; \Delta g = g(\delta) - g(0)$$

$$\frac{\Delta h_\beta}{\delta} = \frac{(p\beta)h_s}{k}$$

where x_β is any N.

In particular in the column of basic unknowns x_α in row α

$$x_\alpha(\delta) = x_\alpha + \frac{(\alpha s)x_p}{k} \cdot \delta$$

and if there is no change of basis

$$\frac{\Delta C}{\delta} = - \frac{x_p h_s}{k} \quad \text{as a generalisation of the simple formula for the derivative used earlier under I.1.}$$

Although the change of basis required cannot be made if $(ps) = 0$ the formulae remain valid since everything is continuous and $(ps) \rightarrow 0$ is permissible. By using the dual the formulae of case 2. apply to case 1. as well.

A change of basis may occur due to two different reasons R_1 and R_2 :

R_1 A value $h_\beta(\delta)$ is negative. In this case x_β enters the basis and the simplex procedure has to be continued accordingly. It may at this stage happen that C becomes unbounded; but this is unlikely in an actual application unless there is a mistake.

R_2 One of the basic variables $x_\alpha(\delta)$ in row α is negative. In this case the corresponding N of the dual enters the basis and the simplex procedure has to be continued on the dual.

Any combination of several R_1 and R_2 may be found; one has to look for the smallest critical values $|\delta_0|$ in both directions in order to find the correct changes of basis to apply.

The particular case of an error in a cost figure is dealt with in available programmes using the shadow prices. An error in a constant term is equivalent to one in a cost figure of the dual.

*) Information from Mr. Beale.

The result of I.2 can be put:

	$x_j \ s_i$	Results affected by δ	Critical value δ_0 for change of basis
1.	B B	only s_i	$\delta_0 = - \frac{s_i}{x_j}$
2.	B N	all h_β β any column of N all x_α α any row of B p row of x_j as a B x_α the B in row α	R_1 : negative $h_\beta(\delta)$ $\delta_0 = \frac{h_\beta}{\begin{vmatrix} h_\beta & h_s \\ (p\beta) & (ps) \end{vmatrix}}$ R_2 : negative $x_\alpha(\delta)$ $\delta_0 = \frac{x_\alpha}{\begin{vmatrix} x_\alpha & x_p \\ (\alpha s) & (ps) \end{vmatrix}}$
3.	N B	none	
4.	N N	only h_j	$\delta_0 = \frac{h_j}{h_s}$

1. is a particular case of 2. R_2

3. is a particular case of 4.

This removes all errors of kind 3. and makes those of kinds 1. and 4. comparatively harmless.
The awkward errors remaining are those of kind 2.

II. Combined Influence of several δ

The main reason for the difficulty of dealing with II is:

Due to the possible change of basis produced by one error the variables change from B to N and vice-versa. Any simple case 1. 4. or even 3. may be changed to the longwinded 2. for further consideration. In a small matrix one may follow up all the possibilities with some effort in any particular case. In a large matrix the effort is likely to become prohibitive quite soon.

If $|\delta|$ is large even I and a small matrix may become a tedious task if the largest $|\delta|$ produces several changes at a time in a case 2.

Suggested Substitutes for a Sensitivity Analysis

It requires a considerable effort on the part of the investigator of each LP model of even moderate (and still more of large) size to construct the matrix and to estimate all the various coefficients required. That is probably the explanation for the many reported cases where the effort has been considered sufficient at that stage and the rest of the investigation has been put to the computer and its dictum has been accepted as the final and indisputable truth.

Since in any large scale model even the solution produced by the computer is far above any possible computation by hand it is obviously impractical to go through an error investigation by hand.

It may, however, be possible to insist on an estimate (however crude) of the maximum size of the error of each coefficient at the time where the coefficient is established.

It is probably useless to hope for more distinctions than:

Either two errors have the same basis and hence have correlations 1 or - 1 respectively
or else two errors are different and "hence" independent.

It is further probably futile to try anything but to assume a normal distribution of each single coefficient error.

Having made these assumptions it is not a problem to call the estimated upper margin of a particular error 2σ and to sample randomly from the assumed normal error distributions.

Changing the matrix correspondingly leads to a new solution. It is suggested to repeat this sampling trial many times and to watch:

Is there (or is there not) a change of results that upsets the impression:

"It does not matter, it is always the same or very much the same".

Depending on what happens in a long series of these repetitions

one will be in a position to decide

either (A): This was a ritual dance, let us give it up.
or (B): It looks reasonable, let us use the result.

All this can be done by computer. Only the results would have to be judged by inspection.

It would be preferable to decide in advance what kind of events in the sampling procedure are to lead to decisions (A) or (B) respectively, but the author cannot imagine a user of a large model who would be prepared to take this kind of decision in advance.

One can establish in a single run each, an upper bound for the possible total change of C in each direction due to errors in all a_{ij} . One only has to use the assumption that C is bounded.

- (1) At each stage δ affects C iff it is a case 2;
and even if the basis changes C is continuous.

$$(2) \quad \frac{\Delta C}{\delta} = - \frac{x_p h_s}{k} \quad k = 1 - (ps) \cdot \delta$$

Since $k > 0$ has been assumed and since $x_p \geq 0$
the sign of ΔC is that of $-h_s \delta$.

i.e. iff $s = s_i$ is really a variable of the model

$$\left. \begin{array}{l} \delta > 0 \text{ decreases } C \\ \delta < 0 \text{ increases } C \end{array} \right\} \text{ since } h_s > 0$$

but iff $e_i = -s_i$ is the real slack the opposite happens
since $h_s < 0$.

Hence C is a monotonic function of each single δ moving in a pre-determined direction common to all δ in the same row. A change of the constant term of this equation works in the opposite direction.

Therefore it is sufficient to choose all δ such that each produces its maximum effect in a common direction, in order to produce an upper and a lower bound for C in two runs of the computer.

If these bounds happen to be sufficiently close together one can safely forget about further error analysis in a particular case.

The opposite statement does not apply - it would be asking too much of a model that it must be safe even if all errors actually happen to work in the same direction.

The author's own practical experience has so far been restricted to a single case where one particular error was so large that already the first attempt of error trial (without any sampling, using the extremes only and neglecting all other errors) led to the clearcut result: Algebraic dance. It was given up. This case actually caused the author's being concerned with sensitivity analysis in LP Models.